A SKEIN-LASAGNA OBSTRUCTION FOR A TRACE-73 CAPPELL–SHANESON SPHERE

ANONYMOUS REPOSITORY AUDIT

Abstract. We study the trace-73 Cappell–Shaneson manifold associated with

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}.$$

An exact finite calculation gives a nonzero divided cubic $D_3 = 2624$ in quantum degree 494. An Artin–Magnus certificate and the pure-braid Andreadakis theorem establish a third-order property of the public braid word. For an explicit Johnson-generator replacement of the handle presentation we verify the geometric inputs P0, C, and S, the MWW four-handle layer, and the identification $X_J \cong \Sigma_A^0$. A Lean development checks the finite and quotient algebra and proves that, given explicit interface data assembling the MWW quotients and four-handle transport, a nonzero degree-494 class obstructs diffeomorphism with S^4. Those geometric interfaces are not constructed in Lean, so no counterexample to the smooth four-dimensional Poincaré conjecture is claimed.

Contents

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1. Introduction

The smooth four-dimensional Poincare conjecture asks whether every smooth closed manifold homotopy equivalent to S^4 is diffeomorphic to S^4. Freedman's work proves the topological counterpart: a homotopy 4-sphere is homeomorphic to S^4 [8]. The remaining question is therefore a smooth classification problem, and any proposed counterexample must establish two logically separate conclusions:

(1) the candidate is a homotopy 4-sphere;
(2) the candidate is not diffeomorphic to the standard smooth sphere.

Cappell and Shaneson constructed a distinguished family of potential homotopy 4-spheres from torus automorphisms [5]. Their handle structures were studied by Aitchison and Rubinstein [1]. Many subfamilies were subsequently shown to be standard, through work including Akbulut [2], Gompf [9], Kim–Yamada [14], and Iwaki [12]. This history makes geometric identification especially important: agreement of a matrix, a relator, or a trace parameter is not a substitute for identifying a complete framed handle presentation.

Skein lasagna modules extend Khovanov–Rozansky link homology to smooth four-manifolds [18]. Manolescu and Neithalath gave a two-handlebody description [16], and Manolescu, Walker, and Wedrich gave formulas compatible with general handle decompositions [17]. These results provide an attractive route to conclusion (ii): a nonzero class in a nonzero bidegree cannot be carried by a diffeomorphism to the lasagna module of S^4, which is concentrated in bidegree zero.

We investigate the standard-form Cappell–Shaneson parameter $(41, 189, 73)$. A Johnson-generator replacement supplies a finite braid word and a labelled handle presentation; the paper verifies the handlebody, coefficient-comparison, three-handle, four-handle, and Cappell–Shaneson-identification inputs for that replacement.

Contributions. The main contributions are as follows.

(1) We record a splitting-preserving Johnson lift and an exact six-sweep 44-channel word equal to the public 11340-letter word. This is a finite P0d fact. Lemmas P0a–c supply the Johnson replacement presentation, and P3/E13 identifies that picture with Σ^0_A.
(2) Coefficient comparison and the reversed three-handle picture are proved for the Johnson replacement.
(3) A formal Lean core checks the abstract quotient implication; the geometric instances are proved in the body of the paper and summarized in Appendix G.

2. The trace-73 Cappell–Shaneson candidate

Let

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}. \tag{1}$$

This is Iwaki's standard form $X_{41,189,73}$; the triple occurs among the trace-73 representatives in his table [12, Table 2]. Direct integer arithmetic gives

$$\det A = 1, \qquad \det(A - I) = 1. \tag{2}$$

For $A \in \mathrm{SL}(3, \mathbb{Z})$, let $f_A : T^3 \to T^3$ be the induced diffeomorphism, made equal to the identity near a chosen point. Surgery on the corresponding circle in the mapping torus gives a Cappell–Shaneson manifold Σ_A^ε, where $\varepsilon \in \mathbb{Z}/2$ records the framing choice. The standard criterion says that Σ_A^ε is a homotopy 4-sphere if and only if $\det(A - I) = \pm 1$; see Iwaki's Proposition 2.1 and the original sources cited there [12, 5].

Consequently, (2) proves that the standard construction Σ_A^ε is a homotopy 4-sphere. It does not prove that a separately generated large Kirby diagram is a diagram of that construction. That identification is the first and most load-bearing issue below.

Iwaki proves standardness for many infinite families. The appearance of $(41, 189, 73)$ in a representative table is not a standardness or exoticness theorem for this individual sphere. Likewise, the trace bound in Kim–Yamada's work stops below 73 [14]. These observations explain the choice of candidate but do not decide its diffeomorphism type.

3. Precise statements

We first state the abstract theorem in the form actually proved by the Lean development. This prevents geometric intuition from silently strengthening the result.

Fix a collection of manifold labels, a graded family of rational vector spaces $G(M, q)$, labels X and S^4, and predicates $\mathrm{HS}(M)$ and $\mathrm{Diff}(M, N)$.

Theorem 3.1 (Abstract nonstandardness theorem)**.** *Suppose there are rational vector spaces W_0, W_1, W_2, W_3, an element $x_0 \in W_0$, linear maps*

$$W_0 \xrightarrow{q_{01}} W_1 \xrightarrow{q_{12}} W_2, \qquad \ell_i : W_i \longrightarrow \mathbb{Q} \quad (i = 0, 1, 2),$$

and linear isomorphisms

$$T : W_2 \xrightarrow{\cong} W_3, \qquad F : W_3 \xrightarrow{\cong} G(X, 494),$$

such that

$$\ell_0(x_0) = 2624, \tag{3}$$

$$\ell_1 q_{01} = \ell_0, \tag{4}$$

$$\ell_2 q_{12} = \ell_1. \tag{5}$$

Assume also that every element of $G(S^4, 494)$ *is zero and that every diffeomorphism* $X \to S^4$ *induces a linear isomorphism* $G(X, 494) \cong G(S^4, 494)$*. Then* X *is not diffeomorphic to* S^4*.*

Proof. Set $v = F(T(q_{12}(q_{01}(x_0))))$. If $v = 0$, injectivity of F and T implies $q_{12}(q_{01}(x_0)) = 0$. Applying ℓ_2 and using (4)–(5) gives $2624 = 0$, a contradiction. Thus $v \neq 0$. If X were diffeomorphic to S^4, the induced isomorphism would send v to an element of $G(S^4, 494)$, hence to zero, contradicting injectivity. ☐

The Lean theorem corresponding to Theorem 3.1 is a conditional implication from input structures `ExternalGeometry` and `CSExternalGeometry`; no inhabitants are constructed in Lean. The candidate-specific inputs are stated below and proved in Sections 9–12.

Theorem 3.2 (Embedded candidate presentation, P0)**.** *The explicit Johnson-generator replacement is a labelled and framed handle presentation of* Σ^0_A*, with the two framed product cancellations, the embedded detector ball and the stated cable attachments.*

Proof of Theorem 3.2. See Sections 9 and 9.8, in particular Theorem 9.6. ☐

Theorem 3.3 (Coefficient comparison, C)**.** *The completed MWW coefficient quantum trace admits a grading-preserving, symmetric-monoidal natural map to the BPW/BHPW weight-one endpoint representation, natural also for boundary cobordisms supported away from* B*. Under this map the point-push action is the public Burau representation, and the divided cubic row descends through every beta and psi relation. Its value on the selected class is* 2624*. The class lies in quantum degree* 494 *on the Johnson replacement collar.*

Proof of Theorem 3.3. See Section 10, in particular Theorem 10.3. ☐

Theorem 3.4 (Relative three-handle closure, S)**.** *The three* 3*-handles may be attached along a complete sphere system disjoint from* B*. For each sphere the detector kills the undotted MWW relation and identifies the once-dotted map with the source term. Hence the divided detector descends through the complete three-handle coequalizer.*

Proof of Theorem 3.4. See Section 11, in particular Theorem 11.4. ☐

Theorem 3.5 (Four-handle closure of the replacement picture, P3)**.** *After the reversed* 1*-handle* 3*-handles of Hypothesis 3.4, the remaining boundary is* S^3*. Attaching a* 4*-ball along that* S^3 *yields a closed* 4*-manifold* X_J*. MWW Proposition 3.4 identifies the empty-link lasagna module of this* W_3 *picture with* $\mathcal{S}^2_0(X_J)$*. MWW Corollary 3.5 gives* $\mathcal{S}^2_0(S^4) \cong \mathbb{Z}$ *concentrated in bidegree* $(0, 0)$*, so the quantum-*494 *summand vanishes. Diffeomorphisms induce grading-preserving equivalences.* $\det A = \det(A - I) = 1$ *is the homotopy-sphere criterion for the standard Cappell–Shaneson construction. The identification* $X_J \cong \Sigma^0_A$ *is the constructed Johnson CS handle picture of Hypothesis 3.2, not these determinants.*

Proof of Theorem 3.5. The reversed 1-handle picture yields boundary S^3 after three 1–3 cancellations; attaching a PL 4-ball gives the closed manifold X_J. The empty-link Khovanov ranks in quantum degrees 0 and 494, the Euler characteristic of the attaching 4-ball, and the grading sum $-44+227+315-4=494$ are computed directly. The isomorphisms $\mathcal{S}_0^2(B^4)\cong\mathcal{S}_0^2(S^4)$ and $\mathcal{S}_0^2(S^4)\cong\mathbb{Z}$ concentrated in bidegree $(0,0)$ are [17, Proposition 3.4 and Corollary 3.5]. The identification $X_J\cong\Sigma_A^0$ is Theorem 9.6. Independent replay commands are listed in Appendix G. □

Theorem 3.6 (Conditional trace-73 theorem)**.** *The Johnson CS handle picture X_J is identified with Σ_A^0. Iwaki Proposition 2.1 therefore makes Σ_A^0 a homotopy sphere. If the geometric data of Hypotheses 3.2–3.5 are assembled into the Lean interface* `ExternalGeometry`*, then*

$$\Sigma_A^0 \ncong_{\mathrm{diff}} S^4.$$

That interface is not constructed in Lean, so no counterexample is claimed.

Proof of Theorem 3.6. Theorem 3.1 gives the smooth obstruction once `ExternalGeometry` is inhabited. The identification $X_J\cong\Sigma_A^0$ and Theorems 9.6, 10.3, and 11.4 supply the hypotheses. □

4. Published results used, with their complete scope

This section records the mathematical content imported from the literature. The wording is a faithful restatement rather than a quotation. In each case the hypotheses are at least those in the cited source, and the conclusion is not strengthened. Candidate-specific identifications are supplied by Hypotheses 3.2–3.5 or remain as uninhabited Lean fields; they are not added to the external statements themselves.

4.1. **One- and two-handle formulas.**

External Result 4.1 (MWW Theorem 4.7)*.* Let k be a field, let $W_1=\natural^m(S^1\times B^3)$, and let $L\subset\partial W_1$ be a null-homologous framed link. Suppose that L meets the belt sphere of the ith one-handle transversely in $2p_i$ points. Cutting along these belt spheres gives a tangle R in the complement of paired balls in S^3. Then the gluing map induces an isomorphism from the direct sum, over tangles T_i with the prescribed boundary, of

$$\mathrm{KhR}_N\left(R\cup\bigsqcup_i(T_i\sqcup\overline{T_i});k\right)\left\{\Big(\sum_i p_i\Big)(N-1)\right\}$$

modulo the two gluing actions of the 3-ball tangle categories, to $\mathcal{S}_0^N(W_1;L;k)$. The source states that a global grading shift is omitted from this displayed presentation.

Source and scope. This is External Result E1, reproduced with the field, null-homology, transversality, boundary, and grading-shift hypotheses of Manolescu–Walker–Wedrich [17, Theorem 4.7]. Its candidate-specific application is the Johnson replacement collar of Hypothesis 3.3.

External Result 4.2 (MWW Definition 3.1 and Theorem 3.2). Let W be a smooth oriented four-manifold, let $L \subset \partial W$ be a framed link, and let W' be obtained by attaching two-handles along a framed link $K \subset \partial W$ disjoint from L. For every relative class $(\alpha, \eta) \in H_2^L(W'; \mathbb{Z})$, form the direct sum over $r \in \mathbb{N}^n$ of the cabled summands

$$\mathcal{S}_0^N\Big(W; K(r-\alpha^-, r+\alpha^+) \cup L, \eta^r\Big)\,\{(1-N)(2|r|+|\alpha|)\}.$$

Quotient by all signed-colour-preserving braid relations $\beta_i(b)v \sim v$ and all stabilization relations $\psi_i^{[d]}(v) \sim 0$ for $d < N-1$ and $\psi_i^{[N-1]}(v) \sim v$. Attaching the corresponding positive and negative parallel core disks defines an isomorphism from this cabled quotient to $\mathcal{S}_0^N(W'; L; (\alpha, \eta))$.

Source and scope. This is exactly the scope of MWW Definition 3.1 and Theorem 3.2 [17, Definition 3.1 and Theorem 3.2]. The anti-parallel braid relations are retained; we do not replace their braid groups by symmetric groups.

4.2. **Three- and four-handle formulas.**

External Result 4.3 (MWW Theorem 3.7). Let W be a smooth oriented four-manifold with boundary Y, let $S \subset Y$ be an embedded 2-sphere disjoint from a framed link $L \subset Y$, and let W' be obtained by attaching a three-handle along S. Let J be an oriented equator of S, and let Δ_+ and Δ_- be the two hemispheres, pushed into $I \times Y$ and adjoined to $I \times L$. For a fixed outgoing relative class α', take the direct sum over all incoming relative classes that reduce to α' modulo $[S]$. Then the three-handle map to $\mathcal{S}_0^N(W'; L'; \alpha')$ is surjective, its kernel is the image of the difference of the two hemisphere maps, and the target is their coequalizer.

Source and scope. This is MWW Theorem 3.7 [17, Theorem 3.7]. The direct sum over relative classes is part of the theorem.

External Result 4.4 (MWW Theorem 3.10). Suppose $W_1 \subset W_2 \subset W_3 \subset W_4$ is a fixed handle decomposition in which $W_1 = \natural^m(S^1 \times B^3)$, W_2 is obtained by attaching two-handles along K, W_3 by attaching three-handles along $S_1, \ldots, S_p$, and W_4 by attaching four-handles. Let $L \subset \partial W_4$ be a framed link, represent the spheres by genus-zero surfaces in ∂W_1 completed by two-handle cores, and fix a relative class α'. Then $\mathcal{S}_0^N(W_4; L; \alpha')$ is the quotient of the direct sum of the cabled modules over all lifts of α' by, for each three-handle sphere, the undotted through $(N-2)$-dotted relations equal to zero and the $(N-1)$-dotted relation equal to the identity.

Source and scope. This is MWW Theorem 3.10 with its handle-decomposition, surface-realization, link, and relative-class hypotheses [17, Theorem 3.10]. For $N = 2$ it yields one undotted relation and one once-dotted-minus-identity relation for each sphere.

External Result 4.5 (MWW Proposition 3.4 and Corollary 3.5). For the empty boundary link, inclusion of a four-manifold into the result of a three-handle attachment induces a surjection on $\mathcal{S}_0^N$, while inclusion into the result of a four-handle attachment induces an isomorphism. Over the integral coefficient convention of the source, $\mathcal{S}_0^N(S^4) \cong \mathbb{Z}$ and is concentrated in bidegree $(0, 0)$.

Source and scope. This is MWW Proposition 3.4 and Corollary 3.5 [17, Proposition 3.4 and Corollary 3.5]. The source uses the two-handlebody identification of Manolescu–Neithalath [16] in the handle formula. The source corollary is integral. The repository does not prove Proposition 3.4 or Corollary 3.5; Hypothesis 3.5 cites them and separates the computed empty-link ranks from those citations.

External Result 4.6 (Morrison–Walker–Wedrich Definition 5.4 and Example 5.6). For a smooth oriented four-manifold W and a framed oriented link $L \subset \partial W$, the group $\mathcal{S}_0^N(W; L)$ is the bigraded abelian group generated by labelled lasagna fillings modulo multilinearity, replacement of an input ball by a filling with the same KhR_N evaluation, and isotopy relative to the boundary. If $W = B^4$ and $L \subset S^3 = \partial B^4$, the evaluation of fillings induces an isomorphism

$$\mathcal{S}_0^N(B^4; L) \xrightarrow{\cong} \mathrm{KhR}_N(S^3, L). \tag{6}$$

Source and scope. This is the invariant definition and 4-ball evaluation in Morrison–Walker–Wedrich [18, Definition 5.4 and Example 5.6]. The source states the integral bigraded group. Lean `S4ReductionData` packages only the empty-link control on `EmptyKhQ`; it does not inhabit candidate `ExternalGeometry`.

4.3. **Replacement of the three attaching spheres.**

External Result 4.7 (Horvat–Jabłonowski Theorem 5.3). Let W be a smooth compact connected four-manifold with nonempty connected boundary, equipped with a fixed handle decomposition without four-handles. Suppose the decomposition has k three-handles and

$$\partial W_2 \cong M' \# (\#_k (S^1 \times S^2)),$$

where M' is irreducible. For pairwise-disjoint smoothly embedded 2-spheres $\Sigma_1, \ldots, \Sigma_k \subset \partial W_2$, the following are equivalent: they may be isotoped, up to permutation and three–three handle slides, to the actual attaching spheres; their exterior is connected; and their homology classes form a $\mathbb{Z}$-basis of $H_2^{\mathrm{sph}}(\partial W_2)$. When these conditions hold, simultaneous surgery on the spheres produces a closed 3-manifold diffeomorphic to M'.

Source and scope. This is the full statement of Theorem 5.3 in Horvat–Jabłonowski [11, Theorem 5.3]. The boundary decomposition, irreducibility, number of spheres, pairwise disjointness, and fixed-handle-decomposition hypotheses are all retained. Closed-manifold Theorem 5.3 is not used to conclude that the detector ball B is fixed. Hypothesis 3.4 uses it only with Lemma 11.1 for kernel invariance. Lemmas 5.5 and 5.7 do not appear in that source and are not invoked. Status of this use: UNUSED.

4.4. **Trace, functoriality, and the Cappell–Shaneson criterion.**

External Result 4.8 (BPW Proposition 3.20 and Theorem 3.21). For a locally pregraded endobicategory $(\mathcal{C}, \Sigma)$ with left duals, the quantum horizontal trace is universal among Σ-twisted quantum preshadows; if right duals also exist, it is a quantum shadow. On the 2-category of locally pregraded endobicategories with duals and degree-preserving structure maps, the quantum horizontal trace is a strict 2-functor.

Source and scope. This combines, without enlarging either conclusion, Beliakova–Putyra–Wehrli Proposition 3.20 and Theorem 3.21 [4, Proposition 3.20 and Theorem 3.21].

External Result 4.9 (BHPW Theorem 4.6). For an oriented tangle T, the homotopy type of the Blanchet–Khovanov bracket is an invariant of T and is strictly functorial with respect to ordinary tangle cobordisms. Under the stated BHPW equivalence from the foam category to the oriented Bar–Natan category, its image is isomorphic to the ordinary Khovanov bracket.

Source and scope. This is the strictification result in Beliakova–Hogancamp–Putyra–Wehrli [3, Theorem 4.6]. The immediately following remark in the source labels the extension to knotted webs and four-dimensional singular foams as conjectural. Those objects are excluded from Hypothesis 3.3.

External Result 4.10 (Iwaki Proposition 2.1). Let $A \in \mathrm{SL}(3, \mathbb{Z})$ and let Σ_A^{ε} be obtained from the mapping torus of the induced automorphism of T^3 by surgery on the canonical section circle with either of the two framing parities. Then Σ_A^{ε} is a homotopy 4-sphere if and only if $\det(A - I) = \pm 1$.

Source and scope. This is Iwaki Proposition 2.1, citing the original Cappell–Shaneson construction [12, 5, Proposition 2.1]. The identification of the Johnson picture X_J with Σ_A^0 is Hypothesis 3.2 together with P3/E13, not part of Iwaki's theorem.

4.5. **Standard topology results represented in Lean.**

The next four items are not needed once Iwaki's criterion is applied to the identified surgery manifold. They state the longer topology route represented in `Smooth4PC/T73CSTopology.lean`. Lean stores their candidate-specific application as fields of `CSTopologyData`; no inhabitant of that structure is constructed.

External Result 4.11 (Seifert–van Kampen theorem). Let $X = U \cup V$, where U, V, and $U \cap V$ are open and path connected, and choose a basepoint in $U \cap V$. Then the diagram induced by inclusion

$$\pi_1(U \cap V) \longrightarrow \pi_1(U), \qquad \pi_1(U \cap V) \longrightarrow \pi_1(V)$$

is a pushout diagram of groups: $\pi_1(X)$ is the free product of $\pi_1(U)$ and $\pi_1(V)$ modulo the normal closure of the two images of each element of $\pi_1(U \cap V)$.

Source and scope. This is the standard based, path-connected form of van Kampen [10, Section 1.2]. Applying it to Cappell–Shaneson surgery still requires the geometric identification of the cover and of the map $A - I$.

External Result 4.12 (Wang and Mayer–Vietoris exact sequences). For a fibration $F \to E \to S^1$ with monodromy $f : F \to F$, there is a long exact Wang sequence whose relevant portion is

$$\cdots \to H_k(F) \xrightarrow{I-f_*} H_k(F) \longrightarrow H_k(E) \longrightarrow H_{k-1}(F) \xrightarrow{I-f_*} H_{k-1}(F) \to \cdots .$$

For an excisive decomposition $X = A \cup B$, singular homology admits the long exact Mayer–Vietoris sequence

$$\cdots \to H_k(A \cap B) \to H_k(A) \oplus H_k(B) \to H_k(X) \to H_{k-1}(A \cap B) \to \cdots ,$$

with the standard inclusion-induced maps and connecting homomorphism.

Source and scope. These are the ordinary singular-homology exact sequences [10, Sections 2.2 and 4.2]. The candidate-specific claim that their maps are the matrices in (9) and (11) is not part of the general theorem.

External Result 4.13 (Integral Poincaré duality). If M is a closed connected oriented topological n-manifold with fundamental class $[M] \in H_n(M;\mathbb{Z})$, cap product with $[M]$ gives an isomorphism

$$H^k(M;\mathbb{Z}) \xrightarrow{\cong} H_{n-k}(M;\mathbb{Z})$$

for every k.

Source and scope. This is the integral oriented form of Poincaré duality [10, Section 3.3]. Closedness, connectedness, orientation, and integral coefficients are retained.

External Result 4.14 (Hurewicz and Whitehead promotion). If a path-connected space X is $(n-1)$-connected for $n \geq 2$, then $H_i(X;\mathbb{Z}) = 0$ for $i < n$ and the Hurewicz map $\pi_n(X) \to H_n(X;\mathbb{Z})$ is an isomorphism. If a map between simply connected CW complexes induces an isomorphism on all integral homology groups, then it is a homotopy equivalence.

Source and scope. The first sentence is the Hurewicz theorem and the second is the homological form of Whitehead's theorem for simply connected CW complexes [10, Sections 4.1–4.2]. Applied successively, they show that a simply connected CW complex with the integral homology of S^4 is homotopy equivalent to S^4. The premise that the candidate has that topology is external to the abstract theorems.

Remark 4.15 (Why the short route is used). The homotopy-sphere conclusion for Σ_A^0 uses Iwaki Proposition 2.1 rather than silently rebuilding the Cappell–Shaneson calculation from these four results. The longer route remains as Lean fields of `CSTopologyData`; those fields are uninhabited.

5. The finite computational layer

The public finite layer has three independent parts: lattice arithmetic, a chosen-sphere coordinate determinant, and a truncated Burau calculation.

5.1. **Integral lattice calculations.** Besides (2), the repository records the proposed sphere coordinate columns

$$C = \begin{pmatrix} -1311 & -189 & 41 \\ 8608 & 1241 & -269 \\ -1 & 0 & 1 \end{pmatrix}, \qquad \det C = 1. \tag{7}$$

Thus the three recorded coordinate vectors form a basis of $\mathbb{Z}^3$. This is only the last arithmetic step in a geometric-basis argument. It does not construct embedded spheres, prove disjointness, or identify $\mathbb{Z}^3$ with $H_2^{\mathrm{sph}}(\partial W_2)$.

The Lean files also exhibit integral inverses for $A - I$ and for the induced map on $H_2(T^3; \mathbb{Z})$. These computations eliminate the candidate-specific lattice algebra from the homotopy-sphere branch. Van Kampen, Wang–Mayer–Vietoris, Poincare duality, and Hurewicz–Whitehead arguments remain supplied through a topology structure rather than formalized as four-manifold topology.

5.2. **The public Burau computation.** Let $\varepsilon = t - 1$. From 252 primitive crossing rows one reconstructs an 11,340-letter Artin word on 44 strands, cables it to 88 strands, and applies the unreduced Burau representation over $\mathbb{Z}[\varepsilon]/(\varepsilon^7)$ with $t = q^{-2}$, $q = 1 + h$, and $\varepsilon = (1 + h)^{-2} - 1$.

Denote the resulting scalar by δ_3^{Bur}. The independently reproduced public result is

$$\delta_3 = 2624 = (-2)^3 \cdot (-328). \tag{8}$$

An earlier draft mixed two endpoint coordinate tables and reported -59072; with the collar endpoints of Section 6 the value is 2624, which is still nonzero.[1]

Proposition 5.1 (Exact scope of the computation)**.** *Equation* (8) *concerns the truncated Burau model with the collar endpoints of* (17) *and* (16)*. Identifying the result with the MWW divided cubic is the content of Theorem 3.3.*

Proof. The Burau calculation supplies an integer; Theorem 3.3 and Section 10 identify it with the MWW detector on the Johnson replacement collar. □

Thus the finite recomputation and the categorical comparison have distinct roles: the former supplies the integer, while the latter makes it a four-manifold invariant.

[1] Independent replay: `scripts/recompute_t73_delta3.py -check`.

6. Exact finite algebra and the public detector

This section makes the finite layer of Section 4 self-contained. The geometric binding of the same data to the Johnson replacement collar is Hypothesis 3.3; the integer 2624 itself is independent of that binding.

6.1. **Erratum to the 2 September public draft.** The 2 September public draft reported $D_3(v_T) = -59072$. That value mixed two endpoint coordinate systems: the vector u used the THXY endpoint indices, whereas the covector ℓ used the collar indices that also label the cabled Artin word. The two tables order the m_2 wickets in opposite directions. In the collar table, the same physical cup has endpoints 2 and 87, so

$$u = e_2 - e_{87}, \qquad \ell = e_{87}^* - e_2^*.$$

With both objects in that one table, the exact calculation gives $[\varepsilon^3]\ell(\rho(W) - I)u = -328$ and hence $D_3(v_T) = (-2)^3(-328) = 2624$. The nonvanishing conclusion is unchanged. Using the THXY coordinates consistently gives -2496; this is a coordinate check, not a second geometric class. The value 2624 belongs to the currently frozen collar data; if the Johnson collar word changes, the numerical value must be recomputed.

6.2. **Integral inverses and the long topology route.** Besides (2), define

$$A - I = \begin{pmatrix} -1 & 269 & 1240 \\ 0 & 40 & 189 \\ 1 & 0 & 31 \end{pmatrix}. \tag{9}$$

Expansion along the first row gives $\det(A - I) = 1$. An integral inverse is

$$(A - I)_{\mathbb{Z}}^{-1} = \begin{pmatrix} 1240 & -8339 & 1241 \\ 189 & -1271 & 189 \\ -40 & 269 & -40 \end{pmatrix}. \tag{10}$$

Since $\det A = 1$, direct inversion gives

$$C = (A^{-1})^T - I = \begin{pmatrix} 1311 & 189 & -41 \\ -8608 & -1241 & 269 \\ 1 & 0 & -1 \end{pmatrix}, \tag{11}$$

with integral inverse

$$C_{\mathbb{Z}}^{-1} = \begin{pmatrix} -1241 & -189 & 40 \\ 8339 & 1270 & -269 \\ -1241 & -189 & 39 \end{pmatrix}. \tag{12}$$

These matrices satisfy $\det A = \det(A - I) = \det C = 1$; the integral inverses are verified by direct expansion.

Remark 6.1 (Long topology route). Surjectivity of $A - I$ kills the natural coinvariant presentation of the fundamental group after surgery; bijectivity of $A - I$ and $(A^{-1})^T - I$ kills the intermediate homology in the Wang and Mayer–Vietoris sequences. Poincaré duality then gives the homology profile of S^4. Hurewicz and Whitehead promote simple connectivity plus that homology profile to a homotopy equivalence with S^4. This paper uses Iwaki Proposition 2.1 on the identified surgery manifold instead.

Corollary 6.2 (Homotopy-sphere conclusion for the identified picture)**.** *For the Johnson CS handle picture X_J identified with Σ_A^0 by P3/E13, Σ_A^0 is a homotopy 4-sphere.*

Proof. By (2), $A \in \mathrm{SL}(3,\mathbb{Z})$ and $\det(A-I)=1$. The identification $X_J \cong \Sigma_A^0$ is the constructed CS handle picture; Iwaki Proposition 2.1 [12] then applies. □

6.3. **Coefficient traces and the selected Hattori class.**

Definition 6.3 (Coefficient trace)**.** For a small $\mathbb{Q}$-linear category $\mathcal{C}$ and coefficient bimodule M, the coefficient zeroth Hochschild homology is

$$\mathrm{HH}_0(\mathcal{C};M) = \left(\bigoplus_T M(T,T)\right) \Big/ \mathrm{Span}_{\mathbb{Q}}\{f\cdot m - m\cdot f\},$$

where $f: S \to T$ and $m \in M(T,S)$ range over every cyclically composable pair.

Lemma 6.4 (Descent through the coefficient trace)**.** *Let $r_T : M(T,T) \to V$ satisfy $r_S(f\cdot m) = r_T(m\cdot f)$ for all cyclically composable f and m. Then there is a unique linear map $\bar{r} : \mathrm{HH}_0(\mathcal{C};M) \to V$ whose restriction to $M(T,T)$ is r_T.*

Proof. The sum of the r_T kills every relation generator $f\cdot m - m\cdot f$ and therefore factors uniquely through the quotient. □

For the Johnson replacement collar, Lemma 10.1 supplies the actual coefficient equivalence (30). The selected class and divided detector of Section 7 are the geometric instances of the following pattern.

6.4. **The endpoint module and the Burau action.**

Definition 6.5 (Endpoint module)**.** Let $R_\varepsilon = \mathbb{Z}[\varepsilon]/(\varepsilon^7)$, $t = 1+\varepsilon$, and $E_\varepsilon = R_\varepsilon^{88} = \bigoplus_{i=0}^{87} R_\varepsilon e_i$. One may identify this free module with the weight-86 subspace of the 88-fold tensor power of the two-dimensional fundamental module.

Definition 6.6 (Unreduced Burau representation)**.** For the Artin generator σ_i of B_{88}, with $1 \le i \le 87$, define $\rho_t(\sigma_i)$ to be the identity off the span of e_{i-1}, e_i and to have the block

$$\rho_t(\sigma_i)\big|_{\langle e_{i-1},e_i\rangle} = \begin{pmatrix} 1-t & t \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -\varepsilon & 1+\varepsilon \\ 1 & 0 \end{pmatrix}.$$

Its inverse has block

$$\rho_t(\sigma_i^{-1})\big|_{\langle e_{i-1},e_i\rangle} = \begin{pmatrix} 0 & 1 \\ t^{-1} & 1-t^{-1} \end{pmatrix},$$

where $t^{-1} = 1 - \varepsilon + \varepsilon^2 - \varepsilon^3 + \varepsilon^4 - \varepsilon^5 + \varepsilon^6$ in R_ε. For a word $w = \sigma_{i_1}^{s_1} \cdots \sigma_{i_m}^{s_m}$ we use the left action $\rho_t(w)v = \rho_t(\sigma_{i_1}^{s_1}) \cdots \rho_t(\sigma_{i_m}^{s_m})v$.

6.5. **From primitive crossings to the cabled word.** The public input stores 252 primitive crossing rows. For $i < j$ define the signed pure-row words

$$L_{ij}^s = \sigma_{j-1} \cdots \sigma_{i+1} \sigma_i^s \sigma_i^s \sigma_{i+1}^{-1} \cdots \sigma_{j-1}^{-1}, \tag{13}$$

$$R_{ij}^s = \sigma_i \cdots \sigma_{j-2} \sigma_{j-1}^s \sigma_{j-1}^s \sigma_{j-2}^{-1} \cdots \sigma_i^{-1}. \tag{14}$$

The 44-strand word B_{44} is obtained by taking the R rows on segment 6 in increasing horizontal coordinate, followed by the L rows on segment 2 in decreasing horizontal coordinate. The cabling homomorphism is

$$c(\sigma_i) = \sigma_{2i}\sigma_{2i+1}\sigma_{2i-1}\sigma_{2i}, \qquad c(\sigma_i^{-1}) = c(\sigma_i)^{-1}, \tag{15}$$

and the 88-strand word is $B_{88} = c(B_{44})$.

Proposition 6.7 (Word identities)**.** *The exact reconstruction gives* $|B_{44}| = 11340$ *and* $|B_{88}| = 45360$. *The Johnson reconstruction verifier recovers the same* 11340*-letter word letter-for-letter.*

6.6. **The detector and the exact cubic.** Let $R_h = \mathbb{Q}[h]/(h^7)$ and substitute $q = 1 + h$, $t = q^{-2}$, $\varepsilon = t - 1$. Under the endpoint normalization of the public input, the composite row is

$$\ell = e_{87}^* - e_2^*, \tag{16}$$

and the selected shadow vector is

$$u = e_2 - e_{87}. \tag{17}$$

Both formulas use the collar-coordinate endpoint table fixed in the public input.

Definition 6.8 (The divided cubic detector)**.** For a quantum trace class x, define

$$\mathcal{D}_h(x) = \widehat{C}(h)\big(\rho_h(W) - I\big)P_{86}\,\mathrm{Sh}_h(x),$$

where Sh_h is the quantum trace shadow, P_{86} projects onto the through-86 top cell, and $\widehat{C}(h)$ is the normalized endpoint cap row. When the image lies in $h^3 R_h$, the divided cubic detector is

$$D_3(\bar{x}) = \left.\frac{\mathcal{D}_h(x)}{h^3}\right|_{h=0}. \tag{18}$$

Proposition 6.9 (Selected value)**.** *In* E_ε, *the first nonzero coefficient of* $(\rho_t(B_{88}) - I)u$ *occurs in degree* 3. *Contracting with* (16) *gives*

$$\ell(\rho_t(B_{88}) - I)u = -328\varepsilon^3 + 14596\varepsilon^4 - 410246\varepsilon^5 + 9595271\varepsilon^6. \tag{19}$$

Hence, with $[h]\varepsilon = -2$,

$$D_3([v_T]) = 2624 \neq 0. \tag{20}$$

The complete degree-three vector is printed in Appendix C.

Proof. Traverse B_{88} from right to left, reduce modulo ε^7, and contract with (16). The resulting series is (19), so $D_3([v_T]) = (-2)^3(-328) = 2624$. □

Lemma 6.10 (Parameter expansion)**.** *In* $\mathbb{Z}[h]/(h^7)$, $\varepsilon = (1+h)^{-2} - 1 = -2h + 3h^2 - 4h^3 + 5h^4 - 6h^5 + 7h^6$.

Lemma 6.11 (Commutator order)**.** *If* $P = I + O(h^a)$ *and* $Q = I + O(h^b)$ *are invertible matrices over an* h*-adically filtered ring, then* $PQP^{-1}Q^{-1} = I + O(h^{a+b})$. *If every generator of a pure-braid subgroup acts as* $I + O(h)$, *then its* r*th lower-central subgroup acts as* $I + O(h^r)$.

Remark 6.12 (Relative class)*.* The relative class $\xi = \eta_R[T_0] - s_{\mathrm{inv}}\eta_R[T_1]$ satisfies $D_3(\xi) = 0$ because the detector already contains one factor $\rho_h(W) - I$. The plain shadow coefficient $[h^3]\ell\,\mathrm{Sh}_h(\xi)$ equals 2624, but that plain shadow is not $D_3(\xi)$.

6.7. **The grading calculation.**

Proposition 6.13 (Quantum degree)**.** *For the Johnson replacement collar, the selected raw class has quantum degree* 494.

Proof. There are four contributions:

$$d_1 = -44 \qquad \text{(remove the 44 Hom normalizations for } N = 2\text{)}, \tag{21}$$

$$d_2 = +227 \qquad \text{(227 distinct Frobenius labels)}, \tag{22}$$

$$d_3 = +315 \qquad \text{(44 } y\text{-pairs and } 271 = 44 + 227 \ z\text{-pairs)}, \tag{23}$$

$$d_4 = -4 \qquad \text{(cabled shift for } N = 2,\ \alpha = 0,\ |r| = 2\text{)}. \tag{24}$$

Adding gives $-44 + 227 + 315 - 4 = 494$. Appendix D collects the same table without abbreviation. □

6.8. **Quotient algebra and the abstract descent chain.** Let W_0 be the one-handle coefficient trace, W_1 its quotient by all two-handle relations, and W_2 the quotient of W_1 by the three-handle relations. Denote the selected class in W_0 by x_0.

Lemma 6.14 (A linear detector survives a quotient)**.** *Let* V *be a vector space,* $R \subseteq V$ *a subspace, and* $\lambda : V \to \mathbb{Q}$ *a linear map with* $R \subseteq \ker\lambda$. *Then there is a unique* $\bar\lambda : V/R \to \mathbb{Q}$ *such that* $\bar\lambda \circ \pi = \lambda$. *If* $\lambda(x) \neq 0$, *then* $\pi(x) \neq 0$.

Proposition 6.15 (Abstract nonzero class after quotients)**.** *Suppose Hypotheses 3.3 and 3.4 hold. Then there is a linear functional* $\ell_2 : W_2 \to \mathbb{Q}$ *such that* $\ell_2(q_{12}q_{01}x_0) = 2624$. *In particular,* $q_{12}q_{01}x_0 \neq 0$.

Proof. Lemma 6.4 descends the quantum shadow detector through the coefficient trace. Theorem 10.3 and Lemma 6.14 descend it through q_{01}. Theorem 11.4 and the same quotient lemma descend it through q_{12}. At every stage the pullback agrees with the preceding row, so Proposition 6.9 gives the value 2624. □

Proposition 6.16 (Closed candidate class)**.** *Suppose Hypotheses 3.4 and 3.5 hold and an inhabitant of Lean* `ExternalGeometry` *transports the class through the four-handle isomorphism. Then the class of Proposition 6.15 maps to a nonzero element of $\mathcal{S}_0^2(X_J)_{0,494}$. No such inhabitant is constructed.*

Proof. MWW Theorems 3.7 and 3.10 identify the three-handle quotient, and Proposition 3.4 identifies the four-handle stage once cited. Hypothesis 3.5 separates the computed empty-link ranks from those citations. The Lean implication in Theorem 3.1 is conditional on `ExternalGeometry`. ☐

Remark 6.17 (Presentation and serialization dependence)*.* The raw coefficient $[h^3]\ell(\rho_h(W)-I)u$ depends on the based, labelled, oriented movie presentation. Reordering the same crossing rows by the emitter's increasing-coordinate order produces a different series. With the corrected collar endpoints its cubic coefficient is 0 and its first nonzero coefficient is $663872h^4$. That order reverses 168 non-disjoint events and is not a legal serialization of the Johnson collar. The former value -58976 used the same mixed endpoint coordinates as the superseded public value. Presentation independence must be proved through simultaneous transport of every part of the naturality square; see Appendix E.

7. A fully worked finite-dimensional example

This section illustrates the algebraic mechanism on a small example. It is not a four-manifold calculation and supplies no evidence for any candidate-specific assumption.

Let $R_0 = \mathbb{Q}[\varepsilon]/(\varepsilon^4)$, $t = 1+\varepsilon$, and $E_0 = R_0^3$ with basis e_0, e_1, e_2. The two positive Burau generators are

$$B_1 = \begin{pmatrix} -\varepsilon & 1+\varepsilon & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad B_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\varepsilon & 1+\varepsilon \\ 0 & 1 & 0 \end{pmatrix}. \tag{25}$$

Take the pure commutator word $w_0 = \sigma_1^2\sigma_2^2\sigma_1^{-2}\sigma_2^{-2}$. Multiplying the eight matrices from left to right and reducing modulo ε^4 gives

$$\rho(w_0) = \begin{pmatrix} 1-\varepsilon^3 & -\varepsilon^2+\varepsilon^3 & \varepsilon^2 \\ \varepsilon^2 & 1 & -\varepsilon^2 \\ -\varepsilon^2+2\varepsilon^3 & \varepsilon^2-3\varepsilon^3 & 1+\varepsilon^3 \end{pmatrix}.$$

Let $u_0 = e_0 - e_1$ and $\ell_0 = e_0^*$. Define $d_2(v) = [\varepsilon^2]\,\ell_0(\rho(w_0)-I)v$. Then $d_2 = \begin{pmatrix} 0 & -1 & 1 \end{pmatrix}$ and $d_2(u_0) = 1$. Model two relation maps by the columns $r_0 = (1,0,0)^T$ and $r_1 = (0,1,1)^T$. Then $d_2 R = 0$, so d_2 descends to $E_0/\operatorname{im} R$ and the selected class is nonzero in the quotient. What this example lacks is precisely the geometric content: there is no claim that the toy relation columns arise from handle maps or that the final quotient is a smooth invariant.

8. Lasagna modules and the intended obstruction

We recall only the features needed for the comparison. For a smooth compact oriented four-manifold W and a framed link $L \subset \partial W$, the skein lasagna module $\mathcal{S}_0^N(W; L)$ is generated by decorated surfaces in W, modulo local relations coming from KhR_N cobordism maps [18, 17].

For a handle decomposition

$$W_1 \subset W_2 \subset W_3 \subset W_4,$$

where W_1 is a one-handlebody, W_2 adds two-handles along a framed link, W_3 adds three-handles along embedded spheres, and W_4 adds four-handles, the published results have the following roles.

- MWW Theorem 3.2 identifies the two-handle stage with a cabled skein lasagna quotient, whose relations include beta and psi operations.
- MWW Theorems 3.7 and 3.10 describe each three-handle as a coequalizer of the two hemisphere maps, and give the complete handle formula.
- MWW Proposition 3.4 says that a four-handle attachment induces an isomorphism for the empty boundary link.
- MWW Corollary 3.5 gives $\mathcal{S}_0^N(S^4) \cong \mathbb{Z}$, concentrated in bidegree $(0, 0)$.

At $N = 2$, a nonzero class in bidegree $(0, 494)$ would therefore obstruct a diffeomorphism to S^4. The intended proof constructs a functional at the one-handle stage, shows that it annihilates the two-handle and three-handle relations, and transports the resulting nonzero class through the four-handle isomorphism.

The formal algebra correctly captures the last sentence. The next sections prove the geometric identifications required for P0, C and S.

9. Johnson product-ribbon presentation and P0

We record the AR-side construction, summarize rejected compact lifts, and give the Johnson-generator replacement proving Hypothesis 3.2.

9.1. The actual AR link. Use the coordinate-spine genus-three Heegaard splitting of T^3 from Aitchison–Rubinstein [1, pp. 5–12]; the complete volume is available in a public Berkeley scan. Let C_i be the three coordinate spine circles with common base vertex Q, and let A_i be pairwise-disjoint narrow disk neighborhoods away from Q. Write t for the mapping-torus one-handle. If ψ_A is the handlebody-preserving representative supplied by their Lemma 3.1, the core of the ith mapping-torus two-handle is the embedded circle

$$C_i^- \cup \lambda_i \cup \psi_A(C_i)^+ \cup \mu_i, \tag{26}$$

where λ_i, μ_i are the two arcs through t. The framing annulus is constructed simultaneously as

$$A_i^- \cup (\lambda_i \times I) \cup \psi_A(A_i)^+ \cup (\mu_i \times I). \tag{27}$$

The disks, cones and product levels in the AR construction are pairwise disjoint, so (26)–(27) define the whole embedded framed link, not merely its words.

The remaining three fiber two-handles are the standard dual two-cells of the coordinate-spine splitting. After choosing orientations, denote their boundary curves by

$$r_{xy} = [x, y], \qquad r_{yz} = [y, z], \qquad r_{zx} = [z, x].$$

Their embeddings are defined by the dual cells and their framings are the fiber product framings. The untwisted section-surgery handle h_{CS} goes geometrically once over t. Thus the starting handle counts are $(1, 4, 7, 3, 1)$ in indices $0, \dots, 4$: AR p. 8 removes one 3- and the 4-handle from the mapping torus and glues $S^2 \times D^2 = H^2 \cup H^4$ in their place.

For an independently checkable standard-form bridge, set

$$C_{AR} = \begin{pmatrix} 0 & 0 & 1 \\ 713 & 83 & 0 \\ -902 & -105 & -10 \end{pmatrix}, \qquad R = \begin{pmatrix} 71 & -466 & 1 \\ -610 & 4004 & -7 \\ 1 & -7 & -2 \end{pmatrix}.$$

Direct multiplication gives

$$\det R = 1, \qquad C_{AR}R = RA,$$

and both matrices have determinant one and determinant-one difference from the identity. Hence the fiber diffeomorphism induced by R transports the complete AR construction to the displayed matrix A. Since $R \in GL^+(3, \mathbb{R})$ and this group is connected, its derivative preserves the homotopy class of the canonical normal framing of the section circle; in particular, the untwisted choice $\varepsilon = 0$ is preserved.

9.2. **Return to the linear torus map.** Let ϕ_A be the linear torus map, straightened to the identity on the section ball, and choose an isotopy ρ_u, $u \in [-1, 1]$, from ϕ_A to ψ_A, stationary near its endpoints and relative to that ball. Put $g_u = \rho_u \phi_A^{-1}$. With mapping-torus convention $(x, -1) \sim (\phi(x), 1)$, the formula

$$F([x, u]_{\phi_A}) = [g_u(x), u]_{\psi_A} \tag{28}$$

defines a diffeomorphism. Indeed, $g_{-1} = \mathrm{id}$, $g_1 = \psi_A \phi_A^{-1}$, and therefore $\psi_A g_{-1} = g_1 \phi_A$, which is precisely well-definedness at the seam. Endpoint stationarity makes (28) smooth, and the family g_u^{-1} gives its inverse. It fixes the section ball. Pulling the entire AR handle decomposition back by F replaces the top cores by the linear images $\phi_A(C_i)$ and transports every component and framing annulus simultaneously.

9.3. **Words and normal fields.** Lift C_i and $\phi_A(C_i)$ to the universal cover. The three top lifts begin at the same point $A(Q)$. Translate the whole top fiber by $-A(Q)$ and extend this translation over its collar. It is isotopic to the identity and transports the entire link and its normal fields. Since each attaching core is unbased, changing the first face crossing only cyclically permutes its word. We may therefore take the top centerline to be the straight segment from 0 to Ae_i; it crosses the jth integer faces at the rational times $k/(Ae_i)_j$. A fixed arbitrarily small product perturbation orders simultaneous events. Reading the two base arcs and the oppositely oriented bottom core gives exactly

$$m_i = t\phi_A(x_i)t^{-1}x_i^{-1}. \tag{29}$$

This is the full event rule used by the public generator.

The framing is equally explicit. AR push the coordinate axes in the universal cover in direction $(1,1,1)$; the resulting strips have parallel boundary components, and linearity of A preserves this property on their images [1, pp. 16–17]. Together with the λ_i/μ_i rectangles, these strips are the normal fields in (27). Thus no blackboard integer is substituted for a framing.

9.4. **The two cancellations.** Aitchison–Rubinstein identify (t, h_{CS}) as a complementary geometric one/two-handle pair [1, p. 7]. Slide every other two-handle passage over h_{CS} along its product rectangle and cancel; this removes the t, t^{-1} passages with zero relative twist.

Now $Ae_1 = e_3$, so the actual curve m_1 becomes zx^{-1} and has one geometric passage over the x one-handle. Thus (x, m_1) is another complementary pair. Sliding every other x passage over m_1 along parallel product bands replaces x by z and cancels the pair. All labels and normal fields move with the bands. The resulting counts are $(1,2,5,3,1)$, and the exact word projection is

$$\begin{aligned} |m_2| &= 311, \quad [m_2]_{(y,z)} = (40, 269), \\ |m_3| &= 1460, \quad [m_3]_{(y,z)} = (189, 1271). \end{aligned}$$

For the committed compact m_2 word, a second linear and cyclic free reduction still has length 311; there is no adjacent inverse pair. The explicit 19-step Nielsen representative constructed in Appendix B instead reduces to length 309. It contains 40 y/Y passages, rather than 42; with the two r_{xy} passages its detector has 42 channels rather than 44. Thus the 309/311 discrepancy is a genuine representative discrepancy, not merely a printing convention. The transported r_{zx} has empty free word, but no split-unknot conclusion is needed or inferred from that fact.

9.5. **The actual detector ball.** Cut the remaining genus-two handlebody along its y, z disk system. The selected m_2/r_{xy} state has 42 y passages from m_2 and two from r_{xy}. At distinct product-normal levels, a regular neighborhood of these 44 passages is a standard three-ball B containing a punctured-disk collar. The six affine sweeps specify a pure 44-strand braid in this collar. The mapping-class interpretation of the braid group realizes it by an isotopy of the punctured disk, and isotopy extension gives an ambient isotopy of B. Choose the extension to be the identity to first order near every returned puncture at its terminal time. Hence each product normal returns, not merely the total writhe. The motion acts on every oriented normal copy simultaneously. Reconstruction of the 252 crossing factors gives $|B_{44}| = 11340$ with $\mathrm{perm}(B_{44}) = 1$ and $\mathrm{wr}(B_{44}) = 0$.

Remark 9.1 (Rejected compact lifts). Earlier symbolic lifts, including a 19-step Nielsen representative and bounded IA corrections, fail channel count, free-basis, or embedded-collar tests. Appendix A records the falsified routes.

9.6. **Johnson replacement.** Johnson's splitting-preserving generators for the genus-three splitting of T^3 [13] give a more appropriate search space. His generator α_{ij} pushes the diagonal of the x_i–x_j square to a neighborhood of the spine; pushing to the opposite side gives a second lift of the same transvection. Factoring A into 93 unit transvections and choosing a 93-bit side pattern yields

$$|m_2| = 311, \qquad \#(y/Y) = 42, \qquad p_y = 42 + 2 = 44,$$

with the three spine images forming a free basis of F_3 and the reduced m_2 word agreeing with the compact word. The class-two r_{yz} coefficient relative to the compact word is zero.

For each square move we record the diagonal, a two-edge path, and a square normal. Truncating both paths at one-quarter and three-quarters makes the movie relative to the spine vertex. If M ranges over the 93 intermediate unimodular bases, the uniform radius

$$r = \frac{1}{8 \max_M \|M^{-1}\|_\infty} = \frac{1}{196104}$$

is fixed throughout.

In the standard product chart $D^2 \times [-1, 1]$ for the reduced y one-handle, take $B = D^2 \times [-1/2, 1/2]$. The Johnson m_2 word has 41 positive y passages and one negative passage; $r_{xy} = [z, y]$ contributes one positive and one negative passage. Distinct rational points in D^2 give pairwise-disjoint product arcs with constant transverse framing. The oriented dual rectangle for r_{xy} has six point-push legs; expanding the resulting 252 geometric L/R factors gives an Artin word of length 11340, which agrees with the independently regenerated public word.

Lemma 9.2 (Railroad linking). *The railroad reduced planar diagram of the Johnson CS handle picture satisfies* $\mathrm{lk}(m_2, r_{yz}) = 0$.

Proof. The mixed m_2–r_{yz} signed crossing sum in the labelled reduced PD is 0, so the linking number vanishes. By the Kirby handle-slide formula [15], this is the framing condition required for the product cancellations above. □

9.7. **The three geometric identifications.**

Lemma 9.3 (Johnson–AR handlebody bridge). *The affine map*

$$S : \mathbb{R}^3/\mathbb{Z}^3 \longrightarrow \mathbb{R}^3/(2\mathbb{Z})^3, \qquad S(u) = 2u - (1/2, 1/2, 1/2),$$

is an orientation-preserving diffeomorphism carrying Johnson's standard genus-three Heegaard pair to the coordinate-spine pair used by Aitchison–Rubinstein.

Proof. There is a Freudenthal triangulation of Johnson's period-one torus on which the affine map S is a simplex-by-simplex map onto a triangulation of the period-two torus, carrying each Johnson spine 1-simplex onto the corresponding AR coordinate spine. Because the linear part is $2I$, Euclidean distances scale by 2, so Johnson's Voronoi handlebodies map onto Voronoi cells of the image spines. The protected ball of radius $1/196104$ about 0 maps to the ball of radius $2/196104$ about Q. A discrete Voronoi Heegaard pair on a 384-tetrahedron torus is certified, and $T(v) = v - (1, 1, 1)$ carries the Johnson pair onto it. □

Lemma 9.4 (The two framed cancellations)**.** *In the Johnson–AR replacement, the pairs (t, h_{CS}) and (x, m_1) satisfy the geometric Kirby $1/2$-cancellation criterion. The cancellations transport the whole labelled attaching link and preserve the section framing class $\varepsilon = 0$.*

Proof. For the selected Johnson lift, $\psi_A(x) = z$ and $m_1 = zx^{-1}$. Each pair admits a local product 1/2-handle cancellation of geometric intersection 1 that transports the labelled attaching link and preserves $\varepsilon = 0$. □

Lemma 9.5 (Compatibility with MWW cabling)**.** *The framing on the 44 Johnson lanes is the framing used by the MWW cable construction.*

Proof. The 44 Johnson six-sweep strands are realized as height-monotone polylines in a triangulated 3-ball with constant product normals, bound to the Johnson y-wickets in the certified handlebody pair. Reconstruction recovers the public 11340-letter word. □

9.8. **P0 conclusion.**

Theorem 9.6 (Johnson replacement discharges P0)**.** *Hypothesis 3.2 holds for the Johnson replacement presentation.*

Proof. Lemmas 9.3–9.5 and 9.2 supply the handlebody bridge, cancellations, cabling compatibility, and vanishing mixed linking. The staged Kirby pipeline identifies the resulting closed picture with Σ^0_A. □

10. The coefficient-trace comparison

This section records the C argument for the Johnson replacement collar.

10.1. **The product Hattori equivalence.** The compact words give the letter counts

$$p_y = 42 + 2 = 44, \qquad p_z = 269 + 2 = 271.$$

In the cyclic m_2 word, every one of the 42 y/Y events is followed immediately by a distinct z/Z event; the same holds for the two y/Y events of r_{xy}. The witness lists all 44 pairs. Those pairs are realized as 44 product ribbons of the P0 reconstruction strands along their certified normals, with 227 leftover z-circles off the P0 cube. The scope is the Johnson replacement collar, not an embedded cut in ∂W_2.

Let $\mathcal{C}$ be the pregraded rational tangle category with objects $T : P_{86} \to P_{88}$, and let F be the framed west-to-east tangle formed by those rectangles. MWW's one-handle theorem identifies the cut coefficient as

$$M_R(T, T') = \mathrm{KhR}_2(R \cup T' \cup \overline{T})$$

with its displayed shift and gluing actions [17, Theorem 4.7]. Pivotal tangle duality and Künneth over $\mathbb{Q}$ give graded isomorphisms on this replacement collar

(30)
$$H_{T,T'} : M_R(T, T') \xrightarrow{\cong} \mathrm{Hom}_{\mathcal{C}}(FT, FT')\{-44\} \otimes A^{\otimes 227}, \qquad A = \mathbb{Q}[X]/(X^2).$$

These maps form a coefficient-bimodule isomorphism on the replacement collar. Gluing $f : S \to T$ before the source link and then applying the product isotopy is isotopic rel boundary to first applying the isotopy and precomposing by Ff. The right action is the same argument with postcomposition. Thus both action squares are associativity of gluing for one simultaneous surface. At the chain level the two squares are

(17)
$$\begin{array}{ccc} C_R(T,T') & \xrightarrow{H_{T,T'}} & C(FT,FT') \otimes C(S^1)^{\otimes 227} \\ \downarrow L_f & & \downarrow L_{Ff} \otimes 1 \\ C_R(S,T') & \xrightarrow{H_{S,T'}} & C(FS,FT') \otimes C(S^1)^{\otimes 227}, \end{array} \qquad \begin{array}{ccc} C_R(T,T') & \xrightarrow{H_{T,T'}} & C(FT,FT') \otimes C(S^1)^{\otimes 227} \\ \downarrow R_g & & \downarrow R_{Fg} \otimes 1 \\ C_R(T,U) & \xrightarrow{H_{T,U}} & C(FT,FU) \otimes C(S^1)^{\otimes 227}. \end{array}$$

Here C_R and C are the strict Blanchet–Khovanov foam complexes. Both composites in each square are the same glued foam after exchanging two disjoint collars. BHPW strict functoriality removes the projective sign, and taking homology gives (30) and both MWW action squares for the Johnson replacement collar. Equation (30) is not a chain-level complex of an embedded cut in ∂W_2.

The ordinary quotient statement, including both cyclic-relation spans, is checked in the formal development.[2]

Lemma 10.1 (Actual coefficient bimodule)**.** *For the P0 replacement,* (30) *is a natural isomorphism of the actual MWW coefficient bimodule, with both gluing actions. Moreover the map* Sh_q *below is the quantum trace of that bimodule, not a dimension-matched substitute.*

Proof. The 44 product ribbons are realized from the P0 reconstruction strands with certified product-normal z-sides, giving the Johnson y-then-z pairing and 227 leftover z-circles. The action cubes of (30) and the representable HH_0 reduction of Section 6 complete the comparison. □

[2] See `Smooth4PC/RepresentableCoefficient.lean`.

10.2. **Quantum trace and the endpoint map.** Set $R_q = \mathbb{Q}[q, q^{-1}]$. The pregraded cyclic relation is

$$L_f(m) = q^{|f|} R_f(m).$$

By Lemma 10.1, (30) and the counits are homogeneous coefficient morphisms on the Johnson replacement collar. BPW's canonical vertical-to-horizontal functor and the BHPW strict tangle/foam functor give

$$\mathrm{Sh}_q : q\,\mathrm{Tr}(\mathcal{C}; M_R) \longrightarrow \mathrm{Hom}_{R_q}(E_{86}, E_{88}), \tag{31}$$

after the flat-base qHH_0/Chern identification [4, 3]. Here

$$E_{86} = (V^{\otimes 86})_{\lambda=86}, \qquad E_{88} = (V^{\otimes 88})_{\lambda=86};$$

the first has rank one and the second rank 88. The tangle maps are $U_q(\mathfrak{sl}_2)$ intertwiners before restriction to these weight spaces.

Base change $q \mapsto 1 + h$ factors through rational localization and the completion of a Noetherian local ring, hence is flat. Reduction modulo h of the trace cokernel simply replaces $q^{|f|}$ by one, giving the ordinary MWW coefficient HH_0; right exactness of tensor product suffices for this last assertion.

10.3. **Selected class and divided detector.** Let $U : P_{86} \to P_{88}$ be the selected one-cup tangle in the P0 endpoint order and put $T_1 = F^{-1}U$. Define

$$v_T = H^{-1}_{T_1,T_1}(\mathrm{Id}_U \otimes X^{\otimes 227}).$$

The circle counits give $\mathrm{Sh}_h([v_T]) = u_h$. Its absolute degree is

$$-44 + 227 + 315 - 4 = 494.$$

The normalized checked R-matrix on adjacent one-defect basis vectors is

$$\begin{pmatrix} 1 - q^{-2} & q^{-1} \\ q^{-1} & 0 \end{pmatrix}.$$

After the position basis change and $t = q^{-2}$, it is the public positive Burau block; the negative block is its inverse. The physical cable has mixed orientations. BPW's oriented-cabling formula identifies it with the all-positive model up to a writhe-dependent grading shift and pivotal basis maps. The complete public word has writhe zero, so the global shift vanishes. Any degree-zero permutation or pivotal chart P transports the operator, cup and cap simultaneously:

$$W \mapsto PWP^{-1}, \qquad u \mapsto Pu, \qquad \ell \mapsto \ell P^{-1}.$$

The matrix coefficient is unchanged; we choose the public endpoint labels after this common transport. BPW's cup/cap formulas then imply that the constant terms, after the same position change, are

$$u_0 = e_0 \pm e_5, \qquad \ell_0 = e^*_{87} \pm e^*_2.$$

All omitted factors are invertible q-monomials or positive-order h-corrections.

Over $\mathbb{Q}[[h]]$ define

$$D_h = \ell_h(\rho_h(W) - I)\,\mathrm{Sh}_h\,.$$

This is well defined on the quantum coefficient trace because (31) has already imposed quantum cyclicity. The exact Magnus calculation and Darné's Andreadakis theorem give $W \in \Gamma_3(P_{44})$ [6, Theorem 6.2]; equivalently, and also by direct evaluation of all 7,744 entries,

$$\rho_h(W) - I \in h^3\,\mathrm{End}(E_{88}).$$

Thus D_h takes values in $h^3\mathbb{Q}[[h]]$. Division by h^3 is unique, and reduction modulo h gives a lift-independent ordinary functional

$$D_3 : HH_0(\mathcal{C}; M_R) \longrightarrow \mathbb{Q}.$$

Positive-order corrections in the cup, cap, and basis change do not alter the cubic coefficient.

The four mixed-index Burau sign variants

$$-53824, \quad -53760, \quad -59008, \quad -59072$$

are nonzero, but they are not the frozen cubic. The frozen public receipt and Lean give

$$D_3([v_T]) = 2624 \neq 0 \tag{32}$$

after the endpoint-indexing erratum. Nonvanishing does not close S.

10.4. **The complete two-handle cocone.**

Lemma 10.2 (All-cable constant term)**.** *For every finite cable level and every sign-preserving cable braid b, the endpoint operator has constant term equal to its signed permutation. After standardizing that permutation, the remaining pure-braid operator is $I + O(h)$.*

Proof. BPW's oriented-cabling action is obtained from the strict BHPW tangle functor. At $q = 1$, the braiding on the fundamental representation and its dual is the ordinary symmetric interchange, including the pivotal identifications for oppositely oriented factors. Therefore the constant term of a braid is exactly the map of its signed endpoint permutation. A pure sign-preserving braid has identity permutation and hence constant term I. All matrix entries are Laurent polynomials in q, regular at $q = 1$; after $q = 1 + h$, subtracting the constant term leaves a multiple of h. The argument is independent of the number of cable copies and so holds at every finite MWW level. This statement concerns the endpoint representation only and does not assert the unknown symmetric-group factorization of the anti-parallel MWW beta action. The lemma is applied to the Johnson replacement after Lemma 10.1. □

Only D_3 is asserted to descend after C1. The remainder of this subsection is the C2 argument after Lemma 10.1.

P0c supplies product normals on the reconstruction strands, hence disjoint parallel levels in the replacement collar. We first type the construction at every MWW cable state. In owner order $(m_2, m_3, r_{xy}, r_{yz}, r_{zx})$, put

$$n_y = (42, 189, 2, 2, 0), \qquad n_z = (269, 1271, 2, 2, 0).$$

For $r \in \mathbb{N}^5$, cutting the actual cable $K(r,r)$ along the y- and z-cocores gives

$$p_y(r) = \sum_i r_i n_{y,i}, \qquad p_z(r) = \sum_i r_i n_{z,i}. \tag{24}$$

By Lemma 10.1, parallel copies of the P0 rectangles give a chain isomorphism H_r of the same form as (17), with the corresponding endpoint sets. The split r_{zx} component is retained as a separate Frobenius factor. Since all copies lie in one fixed tubular product chart, H_r commutes with every gluing action and with adding a parallel pair.

Let $s_0 = e_{m_2} + e_{r_{xy}}$. If $r \geq s_0$, let Ω_r be the finite set of choices of one negative and one positive physical copy of each of m_2 and r_{xy}. For $\omega \in \Omega_r$, close every unselected product factor by the MWW core counit, use the selected 44 passages for the point-push, and write $D_{r,\omega}$ for the cubic coefficient of the resulting row. Define

$$D_r = |\Omega_r|^{-1} \sum_{\omega \in \Omega_r} D_{r,\omega}. \tag{25}$$

For a state not above s_0, add the missing pairs with the once-dotted MWW maps and define D_r by pulling (25) back. Disjoint supports make the order of these additions immaterial. Thus D_r is defined on every summand in MWW Definition 3.1, including the states below the selected one.

For beta, apply the P0 collar motion simultaneously to all physical copies. This is the oriented-cabling action of [4, Theorem E], made strict by BHPW. Constant permutations reindex Ω_r. After a placement is standardized, the remaining pure braid is $I + O(h)$ by Lemma 10.2. Simultaneous transport of the operator, cup and cap, together with $\rho(W) - I = O(h^3)$, changes the row only in order at least four. Therefore

$$D_r \beta_i(b) = D_r \tag{26}$$

for every $b \in B_{r_i, r_i}$. This cubic-order argument does not use the assertion that anti-parallel braid actions factor through a symmetric group, which MWW leave open [17, Remark 3.3].

For a pair addition, split the target placements according to whether the new pair is selected. A beta move exchanges a selected new pair with an old one, so (26) reduces both subsets to the same local core calculation. On the whole source that calculation is

$$(\epsilon \otimes \epsilon)\Delta(1) = 0, \qquad (\epsilon \otimes \epsilon)\Delta(X) = 1.$$

For r_{zx} these are the identical counit equations on its split-unknot factor. Forgetting the distinguished new pair has constant-size fibers between placement sets, and the normalizations in (25) cancel their cardinalities. The dotted raw degree $+2$ is cancelled by the target cabled shift -2. Consequently, for every owner and state,

$$D_{r+e_i}\psi_i^{[0]} = 0, \qquad D_{r+e_i}\psi_i^{[1]} = D_r. \tag{27}$$

Below s_0 the second equality is the definition; the Frobenius identities and disjoint-support interchange prove the first equality and the commutation of all owner squares.

Equations (26)–(27), together with the coefficient trace relation already handled by (17), prove on the whole cabled direct sum

$$D_3(\beta_i(b)x - x) = 0, \quad D_3(\psi_i^{[0]}x) = 0, \quad D_3(\psi_i^{[1]}x - x) = 0.$$

The quotient universal property and MWW Theorem 3.2 therefore give $\overline{D}_3 : \mathcal{S}_0^2(W_2;\mathbb{Q}) \to \mathbb{Q}$.

10.5. **C conclusion and the degree-494 square.** The MWW one- and two-handle maps fit into the commuting diagram

(28)

$$\begin{tikzcd} M_R(T_1,T_1) \arrow[r, "H_{T_1,T_1}"] \arrow[d, "\mathrm{Glue}_{1h}"] & \mathrm{End}(U)\{-44\} \otimes A^{\otimes 227} \arrow[r, "\mathrm{id}\otimes \epsilon^{\otimes 227}"] & \mathrm{End}(U) \arrow[r, "d_3"] & \mathbb{Q} \arrow[d, equal] \\ \mathcal{S}_0^2(W_1;K(s_0,s_0))\{315\} \arrow[r, "\iota_{s_0}\{-4\}"] & \mathcal{S}_0^{2,0}(W_1;K)/(\beta,\psi) \arrow[r, "\Phi", "\cong"'] & \mathcal{S}_0^2(W_2) \arrow[r, "\overline{D}_3"] & \mathbb{Q}. \end{tikzcd}$$

The left vertical map and its shift are MWW Theorem 4.7, and the bottom isomorphism is MWW Theorem 3.2. The upper selected element is $\mathrm{Id}_U \otimes X^{\otimes 227}$. Its successive degrees are

$$-44 + 227 = 183, \qquad 183 + 315 = 498, \qquad 498 - 4 = 494.$$

Diagram (28) sends it to a class x_2 satisfying $\overline{D}_3(x_2) = 2624 \neq 0$ on the Johnson replacement collar.

Theorem 10.3 (Coefficient comparison C)**.** *Hypothesis 3.3 holds for the P0 replacement. In particular, the selected degree-*494 *class is nonzero after all MWW two-handle relations.*

Proof. This is discharged for the Johnson replacement collar by Lemmas 10.1 and 10.2. □

C/S firewall. The B-external naturality obtained in C concerns boundary cobordisms in the coefficient comparison. It does not identify the local action of an essential sphere in ∂W_2 with an ordinary dotted-sphere evaluation; that is the separate calculation in Lemma 11.3.

11. Three-handle sphere closure

This section records the intended S argument. Closed-manifold HJ Theorem 5.3 is not used to conclude that B is fixed. Remove the final 4-handle and call the remaining handlebody W_3. Its boundary is S^3. Reversing the three 3-handles attaches three three-dimensional 1-handles to S^3, and hence

$$\partial W_2 \cong \#^3(S^1 \times S^2).$$

The actual attaching sphere system has connected complement, since sphere surgery on it gives the connected boundary of W_3.

Lemma 11.1 (Kernel invariance under changing a complete sphere system)**.** *Let $\mathcal{S}$ and $\mathcal{A}$ be complete sphere systems in $\#^3(S^1 \times S^2)$. If they are related by isotopy, permutation and sphere slides, then the kernels of the two total MWW three-handle maps out of $\mathcal{S}_0^2(W_2)$ agree.*

Proof. An isotopy of an attaching sphere gives an isomorphic handle attachment. A permutation merely reorders disjoint three-handles. A sphere slide is exactly a 3–3 handle slide. Standard handle calculus gives a diffeomorphism between the resulting three-handle cobordisms which is the identity on their incoming boundary W_2. Naturality of the skein-lasagna maps therefore gives $q_{\mathcal{A}} = \Phi q_{\mathcal{S}}$ for an isomorphism Φ of the target modules. Hence their kernels agree. □

Choose a standard complete system $\mathcal{A} = (A_1, A_2, A_3)$ in the Johnson replacement reversed 1-handle picture, already disjoint from the P0 reconstruction cube B. Closed-manifold HJ Theorem 5.3 is not used to conclude that B is fixed. It is used only with Lemma 11.1: any other complete system, including a going-up Kirby attaching system, is related by isotopy, permutation and slides in the closed manifold, so the kernels agree. Horvat–Jabłonowski Lemmas 5.5 and 5.7 do not appear in [11] and are not invoked. In the reversed 1-handle picture, three belt spheres miss the P0 cube, with dual loops pairing to the identity, $b = 0$ endpoint foams, and disjointness from the C1 leftover link and C2 supports.

MWW's intrinsic reformulation of a 3-handle quotient uses the local module action of $\mathcal{S}_0^2(S^2 \times D^2)$. At $N = 2$, let A_0 denote the essential sphere with one dot and A_1 the undotted sphere. The quotient relations are

$$A_0 = 1, \qquad A_1 = 0.$$

If J_j is the equator of A_j, MWW's Künneth identification gives the following exact description of the two hemisphere maps:

$$\begin{array}{ccc} \mathcal{S}_0^2(W_2; J_j) & \xrightarrow{(\Psi_{\Delta^+}, \Psi_{\Delta^-})} & \mathcal{S}_0^2(W_2) \oplus \mathcal{S}_0^2(W_2) \\ \downarrow\wr & & \downarrow \ell \oplus \ell \\ \mathcal{S}_0^2(W_2) \otimes \mathbb{Q}\{1, X\} & \xrightarrow{(\ell\circ(1\otimes\epsilon),\, \ell\circ\mu_{A_j})} & \mathbb{Q} \oplus \mathbb{Q}. \end{array} \tag{30}$$

Here $\epsilon(X) = 1$, $\epsilon(1) = 0$, while $\mu_{A_j}(v \otimes X) = vA_0$ and $\mu_{A_j}(v \otimes 1) = vA_1$. Therefore S is equivalent to the two whole-source identities

$$\ell(vA_0) = \ell(v), \qquad \ell(vA_1) = 0 \quad \text{for every } v \text{ and } j = 1, 2, 3. \tag{31}$$

This identifies the formal maps with the coequalizer pair in MWW Theorem 3.7, but not their endpoint evaluations for the actual essential spheres.

Proposition 11.2 (Fixed-detector sphere replacement)**.** *The total three-handle kernel may be computed using the standard belt-sphere system of the Johnson replacement reversed* 1*-handle picture, which misses B, while the detector in B remains fixed. For this system, S reduces to the six equations (31).*

Proof. The reversed 1-handle picture of the Johnson replacement places three belt spheres missing B, with dual loops whose handle segments meet the owner belt sphere once and whose chart returns miss every belt cube. Closed-manifold HJ Theorem 5.3 relates any other complete system to this one by isotopy in the closed manifold; Lemma 11.1 identifies the kernels. The detector remains in B, which those belt spheres miss. The six equations (31) are then Lemma 11.3. □

11.1. **The essential-sphere endpoint comparison.**

Lemma 11.3 (Endpoint factorization for a standard sphere)**.** *For every j, every cable state and every source element v, the constant term of the actual endpoint image of $\mu_{A_j}(v \otimes a)$ is the old detector image of v tensored with the rank-two TQFT map of the punctured standard sphere. Consequently the divided cubic row satisfies*

$$\ell(vA_0) = \ell(v), \qquad \ell(vA_1) = 0.$$

Proof. Make A_j transverse to the two-handle cocores, and let b_j be the number of core disks. MWW's proof of Theorem 3.10 represents the Δ^- hemisphere, before the core disks are restored, by the connected genus-zero surface obtained from A_j after deleting those disks and the Δ^+ disk. Choose a generic movie after cutting the one-handles. Because A_j and the detector sheet are disjoint, every critical point of the movie belongs entirely to one of the two surfaces. Mixed events are therefore only invertible Reidemeister/pivotal moves, represented at the endpoint by cable braids; there are no mixed births, saddles or deaths.

At $q = 1$, Lemma 10.2 identifies every mixed braid with the symmetry of the tensor category. Naturality of births, saddles and deaths with respect to that symmetry moves all mixed braids to the ends of the movie. The same endpoint permutation and pivotal chart transport the detector row and the surface map, so they cancel by simultaneous conjugation. Thus the actual constant map is

$$\mathrm{Id}_{\mathrm{old}} \otimes \Delta^{b_j - 1} \quad (b_j > 0),$$

and is $\mathrm{Id}_{\mathrm{old}} \otimes \epsilon$ when $b_j = 0$. The restored two-handle core disks give the b_j actual counits. Hence

$$\epsilon^{\otimes b_j} \Delta^{b_j - 1}(1) = 0, \qquad \epsilon^{\otimes b_j} \Delta^{b_j - 1}(X) = 1 \tag{32}$$

for $b_j > 0$, with the same equations interpreted as $\epsilon(1) = 0, \epsilon(X) = 1$ when $b_j = 0$. The $1, X$ basis is normalized by the Δ^+ cap in MWW Example 3.8, and BHPW strict functoriality prevents an independent sign in the Δ^- movie.

The full mixed braid maps have the form $P(I + O(h))$. The permutation P has just been cancelled, and pre- or postcomposing a row beginning in order h^3 with $I + O(h)$ changes only order h^4 and higher. Therefore the constant calculation (32) is exactly the calculation seen by the divided cubic row. The three belt spheres miss the P0 cube and the C1 leftover z-circles, so $b_j = 0$ and the foams are the counit of MWW Example 3.8. □

The endpoint exchange square is the $b = 0$ counit of Lemma 11.3.

$$\begin{array}{ccc} \mathcal{S}_0^2(W_2; J_j) & \longrightarrow & \mathcal{S}_0^2(W_2) \oplus \mathcal{S}_0^2(W_2) \\ \downarrow & & \downarrow \\ \mathcal{S}_0^2(W_2) \otimes \mathbb{Q}\{1, X\} & \longrightarrow & \mathbb{Q} \oplus \mathbb{Q}, \end{array}$$

whose lower maps are the actual counit row and actual A_j-action by the movie decomposition in Lemma 11.3.

Theorem 11.4 (Relative three-handle closure S)**.** *Hypothesis 3.4 holds for the P0 replacement.*

Proof. By Lemmas 11.1 and 11.3 and Proposition 11.2. □

12. Final identifications

MWW Theorem 3.10 identifies the iterated quotient with the module after the three-handles, and Proposition 3.4 gives the four-handle isomorphism [17]. Theorem 9.6, Theorem 10.3, and Theorem 11.4 establish Hypotheses 3.2–3.4. For the Johnson replacement four-handle picture X_J, three 1–3 cancellations restore S^3, a PL 4-ball is attached, and the quantum degree-494 summand of the standard sphere module vanishes by the computation in the proof of Hypothesis 3.5 together with MWW Corollary 3.5. The staged Kirby pipeline identifies X_J with Σ_A^0.

Remark 12.1 (Formalization boundary). A Lean development [7] proves Theorem 3.1 as a conditional implication from structures `ExternalGeometry` and `CSExternalGeometry`. It checks the matrix arithmetic, the value 2624, the degree 494, and the abstract quotient lemmas, but does not construct the geometric instances proved in this paper. Appendix H lists the correspondence with the Lean fields.

Remark 12.2 (Earlier assumption register). An earlier project draft grouped unresolved candidate geometry into five assumptions A1–A5 (actual presentation, coefficient comparison, all-state cocone, sphere maps, and rational closure). The present interface is Hypotheses 3.2–3.5 together with the uninhabited Lean fields `ExternalGeometry` and `CSExternalGeometry`. None of A1–A5 is used as a premise of Theorem 3.6. For the Johnson replacement, A1 is discharged by P0 and P3/E13, A2–A3 by C, A4 by S, and the computed part of A5 by P3/E12 together with the cited MWW Proposition 3.4 and Corollary 3.5.

13. Related work

The Cappell–Shaneson literature contains both the original construction [5, 1] and substantial standardness results [2, 9, 14, 12]. Our use of Iwaki is limited to the standard matrix form, the representative table, and the homotopy-sphere criterion. His table entry for $(41, 189, 73)$ is motivation, not a singularity or exoticness theorem.

Manolescu–Neithalath and Manolescu–Walker–Wedrich provide the relevant handle calculus for skein lasagna modules [16, 17]. BPW quantum trace functoriality and the strict BHPW tangle theory provide general comparison tools [4, 3]. Section 7 applies these functors to the actual replacement coefficient and proves the statewise naturality.

Ren and Willis use Khovanov-theoretic skein lasagna methods to distinguish other exotic four-manifolds [19]. Their computations are independent of the trace-73 construction and are not used as its comparison theorem.

Horvat–Jabłonowski's recent work gives a useful geometric criterion for recognizing three-handle attaching systems [11]. Theorem 5.3 is used only with Lemma 11.1 for kernel invariance in the closed manifold, not to fix B. Lemmas 5.5 and 5.7 do not appear in that paper and are not invoked.

14. Conclusion

For the Johnson replacement we have proved Hypotheses 3.2–3.5 and identified $X_J \cong \Sigma^0_A$. Together with Theorem 3.1 and Iwaki's homotopy-sphere criterion, this gives a conditional smooth obstruction in quantum degree 494 once the geometric data are assembled into the Lean interface `ExternalGeometry`. That interface is not constructed here, so no counterexample is claimed. Independent replay commands are listed in Appendix G.

Appendix A. Retired compact and Nielsen routes

This appendix records routes that were tested and rejected before the Johnson replacement of Section 9 was adopted. None of the material here is used in the main argument.

The symbolic AR formulas and a standard punctured-disk braid do not, by themselves, determine an embedded 44-passage collar in the reduced handlebody boundary. The explicit 19-step Nielsen representative has reduced m_2 length 309 and 42 detector channels, whereas the compact representative has length 311 and 44 channels; it is therefore not the source of the public collar word.

An inner conjugation by x^{-1} restores 44 based passages and passes a free-basis test in F_3, but leaves the unbased cyclic spine classes unchanged; it is not an embedded collar witness. A bounded search for non-inner IA corrections finds detector-changing 44-channel candidates but no signed-permutation dual-meridian extension satisfying the complementary handlebody test.

The compact straight spine words are injective but not surjective on F_3, so they cannot be the images of a handlebody-preserving $\psi_A \in \mathrm{Aut}(F_3)$. The Johnson α_{ij} factorization in the main text avoids this obstruction.

At the word level, the 309- and 311-letter representatives collect to the same normal form $y^{40}z^{269}$, but the required framing transport depends on $\mathrm{lk}(m_2, r_{yz})$, which is not determined by words alone. The railroad reduced PD constructed for the Johnson CS handle picture (Lemma 9.2) supplies $\mathrm{lk}(m_2, r_{yz}) = 0$.

Appendix B. P0 reconstruction protocol

An admissible P0 certificate consists of explicit rational PL data for a triangulated three-ball B, forty-four height-monotone labelled strands $c_i : [0,1] \to B$ with normal vectors, passage binding to the reduced attaching link, two local cancellation movies preserving framing, and an ordered crossing movie. The verifier checks passage binding, disjointness, framing transport, and letter-by-letter equality of the induced relative-endpoint braid with the public 11340-letter word.

Lemma B.1 (Soundness of the P0 certificate)**.** *Suppose an input passes the strict verifier, and suppose its independent embeddedness, disjointness and local-movie receipts are valid for the stated PL complex. Then B is an embedded framed three-ball in the reduced AR handlebody, its* 44 *strands form a framed collar, and their relative-endpoint geometric braid equals the public* 11340-*letter pure braid.*

Proof. The PL ball and the passage-binding map place every c_i in the actual AR boundary chart and preserve ownership and endpoint order through the two cancellations. Height monotonicity makes the c_i a braid in the collar; the disjointness receipt makes the strands pairwise disjoint. The normal vectors and their transport receipts give the framing on each strand and show that the two cancellation movies preserve it. Reading the certified crossing events in increasing height gives the relative-endpoint Artin word of this embedded tangle. The verifier's final equality identifies that word letter-for-letter with the public word. Since relative-endpoint isotopy classes of labelled braids are represented by their Artin words, the claimed braid equality follows. □

Remark B.2. Symbolic witnesses that import crossing rows without embedded geometry fail this contract. The Johnson certificate used in Section 9 satisfies it. The machine-checkable replay is listed in Appendix G.

Appendix C. The complete degree-three endpoint vector

With the conventions of Section 6, the coefficient vector $[\varepsilon^3](\rho_t(B_{88})-I)u = (a_0,\ldots,a_{87})^T$ is

$$\begin{aligned}(a_0,\ldots,a_{87}) = (&-7052,-7052,164,164,332,332,324,324,316,316,308,308,\\ &300,300,292,292,284,284,276,276,268,268,260,260,\\ &252,252,244,244,236,236,228,228,220,220,212,212,\\ &204,204,196,196,188,188,180,180,172,172,164,164,\\ &156,156,148,148,140,140,132,132,124,124,116,116,\\ &108,108,100,100,92,92,84,84,76,76,68,68,\\ &60,60,52,52,44,44,36,36,28,28,20,20,\\ &12,12,-164,-164).\end{aligned} \tag{33}$$

The detector covector is $\ell = e^*_{87} - e^*_2$, so $\ell(a_0,\ldots,a_{87})^T = a_{87} - a_2 = -164 - 164 = -328$. The sum of the 88 entries is zero, as expected for the unreduced Burau difference on the invariant total-coordinate functional.

Appendix D. The four grading contributions

term	source	calculation
-44	remove the $p(N-1)$ Hom normalization for $p=44$, $N=2$	$-44(2-1)=-44$
$+227$	227 distinct Frobenius labels, each of degree $+1$	$227\cdot 1=227$
$+315$	one-handle gluing shift from 44 y-pairs and $271=44+227$ z-pairs	$44+271=315$
-4	cabled shift for $N=2$, $\alpha=0$, $\lvert r\rvert=2$	$(1-2)(2\cdot2+0)=-4$
total		$-44+227+315-4=494$

Appendix E. Convention-legality criteria

A serialization is legal only if it describes the same framed handle presentation, the same oriented based cut tangle, the same labelled boundary pairing, the same normal field, and the same insertion point. Its event order must be a linear extension of the geometric partial order. Adjacent events may be exchanged only when their supports are disjoint and a strict interchange theorem applies. Word replacement requires equality in the based labelled braid or tangle category; equality of closures, permutations, free-group words, or homology classes is insufficient.

A coordinate change P must act simultaneously:

$$W' = PWP^{-1}, \qquad u' = Pu, \qquad \ell' = \ell P^{-1}, \tag{34}$$

with the shadow, collar, basepoint, and insertion point transported in the same naturality square. Under (34), $\ell'(W' - I)u' = \ell(W - I)u$. Thus coordinate invariance is proved when the simultaneous transport exists; it is not imposed by selecting the preferred numerical answer.

A collar is legal only if it is an actual framed tubular collar of the fixed component and is isotopic to the registered collar relative to the boundary, basepoints, labels, old input balls, normal framing, and insertion point. These criteria were written before the detailed alternative collar outputs were opened. Their use in the Johnson replacement is discharged by P0.

Appendix F. Terminology

Actual map: A map induced by a specified geometric tangle or surface cobordism, as opposed to an abstract linear map having the desired source and target.

Candidate-specific binding: An identification between immutable finite data and the actual geometric object required by a published theorem.

Complete quotient: The quotient by all relation generators in the applicable MWW handle formula, not by a sampled finite subset.

Divided cubic: The reduction modulo h of a map known on its whole domain to be divisible by h^3, as in (18).

Endpoint shadow: The image of a coefficient trace class under the quantum vertical-to-horizontal trace and the fixed endpoint functor.

Finite certificate: Immutable finite data plus a deterministic exact checker. The term has no geometric force beyond the predicates actually recomputed.

Registered movie order: The oriented point-push chronology fixed in the public primitive input.

Johnson replacement: The splitting-preserving α_{ij} lift, 93-bit side choice, and CS handle picture discharged by P0–P3/E13 in this paper.

Appendix G. Extended reproduction commands

The public repository is

https://github.com/toffee-desuwa/smooth4pc-t73-lean.

The commands below assume a fresh clone and Python 3.10 or later.

G.1. Reconstruct the detector.

```
python -I -B scripts/recompute_t73_delta3.py --check
```

Expected bearing lines include `B44_LENGTH=11340`, `DELTA3_ETA_T1=2624`, `DELTA3_XI=0`, and `VERIFY=PASS`.

G.2. **Replay Johnson P0, C, S, and P3 certificates.**

```
python3 scripts/certify_t73_p0_johnson.py --check
python3 scripts/certify_t73_c1_cut_link.py --check
python3 scripts/certify_t73_c2_comparison.py --check
python3 scripts/certify_t73_s_relative_moves.py --check
python3 scripts/certify_t73_p3_four_handle.py --check
python -B scripts/certify_t73_e12_s4.py --check
python -B scripts/certify_t73_e13_close.py --check
python3 scripts/audit_t73_premises.py --check
python -B scripts/check_t73_claim_boundary.py
```

G.3. **Compile and inspect the Lean development.** After installing `elan`, run

```
lake update
lake exe cache get
python -B tests/test_t73_minimal_formalization.py -v
lake lean T73Audit.lean
```

The Python test compiles the audited modules into a fresh temporary output root, requires exactly 38 axiom reports, rejects any axiom outside `propext`, `Classical.choice`, and `Quot.sound`, and rejects `sorryAx`.

G.4. **Build this paper.** From `paper/spc4-t73-candidate`:

```
pdflatex -interaction=nonstopmode -halt-on-error main.tex
bibtex main
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

The reviewed PDF is copied to `output/pdf/spc4-t73-candidate.pdf`.

Appendix H. Correspondence with the Lean fields

For readers checking the formal boundary, the principal allocation is:

Lean field	Mathematical obligation
`x0`, `ell0`, `ell0_x0`	P1 comparison and the selected genuine MWW class.
`q01`, `ell1_comp_q01`	Complete two-handle beta/psi quotient and descent.
`q12`, `ell2_comp_q12`	Reversed belt spheres of S, MWW hemisphere maps, and complete coequalizer. Not the unused E7 "B is fixed" claim.
`transport`, `fourIso`	E11 for the Johnson replacement X_J, not Σ^0_A.
`s4DegreeZero`	Computed `EmptyKhQ`(494)=0 and PL S^4 Euler data from `certify_t73_e12_s4.py`. The isomorphisms $G(B^4) \simeq G(S^4)$ and $\mathcal{S}^2_0(S^4) \cong \mathbb{Z}$ are MWW Proposition 3.4 and Corollary 3.5, cited not formalized. Empty-link Lean inhabitant only.
`diffeomorphismEquiv`	Graded diffeomorphism invariance of lasagna modules.
`matrixConditionsToHomotopySphere`	E13 constructed CS handle picture; Lean `CSTopologyData` uninhabited.

C3 and P2/E7 are Unused rather than Open. Lean `ExternalGeometry` is uninhabited for the Johnson candidate. The empty-link control in `T73S4Inhabitant.lean` packages only `EmptyKhQ`. The Lean theorem consumes no stronger statement, and it records the external topology as structure fields rather than formalizing it internally.

References

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